String Theory and M-Theory

Lecture 1: From Hadronic Regge Trajectories to Light-Cone Strings

Lectures by Leonard Susskind

Transcript-derived companion notes curated by LazyingArt LLC
Produced with Video2Book

Chapter 1

From Hadronic Regge Trajectories to Light-Cone Strings

Overview. String theory did not begin as a theory of quantum gravity. It began as an attempt to understand the spectrum and scattering of hadrons. Straight Regge trajectories suggested an extended object; duality in pion–pion scattering suggested that two apparently distinct processes were two slicings of one surface. We shall follow those clues into the world-sheet picture, reinterpret the hadronic string as a QCD flux tube, and then use the infinite-momentum frame to derive the transverse mechanics of an open string. The central result is that its effective internal Hamiltonian measures mass squared. A Hooke-law energy proportional to L^2 in that description therefore becomes a rest energy proportional to L, the defining behavior of a string with constant tension.

1.1 The hadronic origin of string theory

Around 1968–1970, the problem that led to strings concerned protons, neutrons, and especially mesons. The quark model was already a serious idea, but the quark content of hadrons had not yet been experimentally established in the modern sense. Gluons were even less prominent in ordinary discussion. Nambu had proposed related gauge-field ideas, but the settled language of quantum chromodynamics did not yet exist. The raw material was instead an unexpectedly rich spectrum of strongly interacting particles.

It is useful to ignore the proton–neutron distinction temporarily and call both states the nucleon. Beyond the nucleon were heavier states with larger spin, followed by still heavier states with still larger spin. Chew and Frautschi organized such families on a plot with angular momentum on the vertical axis and mass squared on the horizontal axis. In the historical units of the period the proton mass was about one BeV; the same unit is now called a GeV.

Measure angular momentum in units of $\hbar$. For a baryon family, the dimensionless spin values are half-integral,

$$J = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots, \tag{1.1}$$

while meson families have integral spin. The remarkable empirical fact was that either family lay approximately on a straight line when plotted against M^2. In modern notation one writes

$$J \simeq \alpha_0 + \alpha' M^2, \tag{1.2}$$

where α_0 is an intercept and α' is the Regge slope. [1]

The original straight line was not compelling because it passed through only two known points. Any two points determine a line. The surprise came later: newly discovered states continued to land close to the same line. Meson trajectories showed the same organization. The pion begins near $J = 0$ with a small mass on the GeV scale; the rho begins at $J = 1$; further excitations continue upward. Five, six, or seven states could appear in a family. Even more strikingly, different baryon and meson trajectories were nearly parallel. Thus an increase by one quantum of angular momentum corresponded to approximately the same increase in mass squared:

$$\Delta J = 1 \quad \Longrightarrow \quad \Delta M^2 \simeq \frac{1}{\alpha'}. \tag{1.3}$$

Question from the audience. What does it mean for different Regge trajectories to be parallel?[a]

Response. Choose one family of particles and plot all of its excited states; they lie along one line. Repeat this for another meson or baryon family. The lines have approximately the same slope even though their intercepts differ. The same increment of angular momentum therefore costs approximately the same increment of M^2. This common slope was called the universal Regge slope.

[a] Lecture timestamp: 00:09:32.

The comparison with ordinary extended objects shows why the pattern mattered. A basketball acquires rotational energy as it is spun, but its trajectory is curved rather than linear, and at sufficiently high angular velocity the ball is torn apart. An atom also has rotational excitations, but eventually it is ionized; its trajectory ends. Hadrons displayed a much simpler sequence and, within the available data, no sign that the sequence was giving out.

This spectrum said that a hadron was not behaving like a point. A point has no spatial structure to spin or deform. An electron has no observed tower of higher-spin internal states and is correspondingly pointlike within present experimental bounds. A hadron, by contrast, can be spun up. Its spectrum suggests compositeness, and the relation between spin and mass hints that the object deforms as its angular momentum increases.

1.2 Duality in pion–pion scattering

Spectroscopy supplied one clue. Pion–pion scattering supplied another. Two pions can combine into a rho-like resonance and later emerge again as two pions. In a conventional Feynman diagram this is an annihilation and re-creation process, customarily called an s-channel process. Turning the same elementary diagram sideways gives a crossed, or t-channel, process in which a rho-like particle is exchanged between the pions.

[1] Editorial clarification: The lecture emphasizes the observed linear relation and its nearly universal slope. The intercept notation is supplied only to state that relation compactly; the trajectories are empirical and approximate.

The rho is not alone. It belongs to a Regge family, so consistency appears to require the rho and all of its excited partners in either channel. Schematically,

$$\mathcal{A}(s,t) \simeq \sum_n \mathcal{A}^{(s)}_{\rho_n}(s,t), \qquad \mathcal{A}(s,t) \simeq \sum_n \mathcal{A}^{(t)}_{\rho_n}(s,t). \tag{1.4}$$

These expressions are not two approximations that should simply be added. Numerical studies found that summing the tower of resonances in the annihilation channel already gave approximately the contribution expected from the exchange channel. Conversely, the exchanged tower could reproduce the resonance description. Adding both complete towers therefore appeared to count the same physics twice. Ordinary perturbative bookkeeping says that if two diagrams exist, both must be included. The hadronic amplitudes instead suggested that the two channel descriptions were dual.

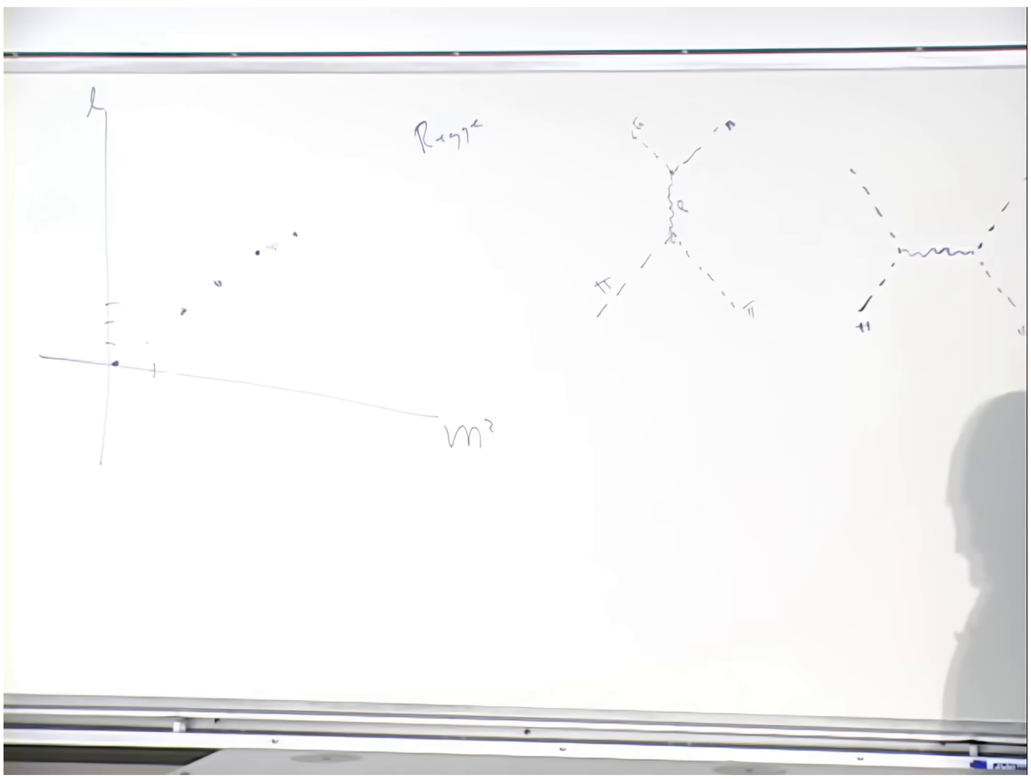

Figure 1.1: The board joins the Chew–Frautschi plot to the annihilation and crossed-channel diagrams, making the spectral and scattering clues part of one problem.

Lecture frame: 00:14:20.

The same relations can be separated into a clean schematic. The first panel records the linear spectrum; the other two represent the alternative channel expansions of the same amplitude.

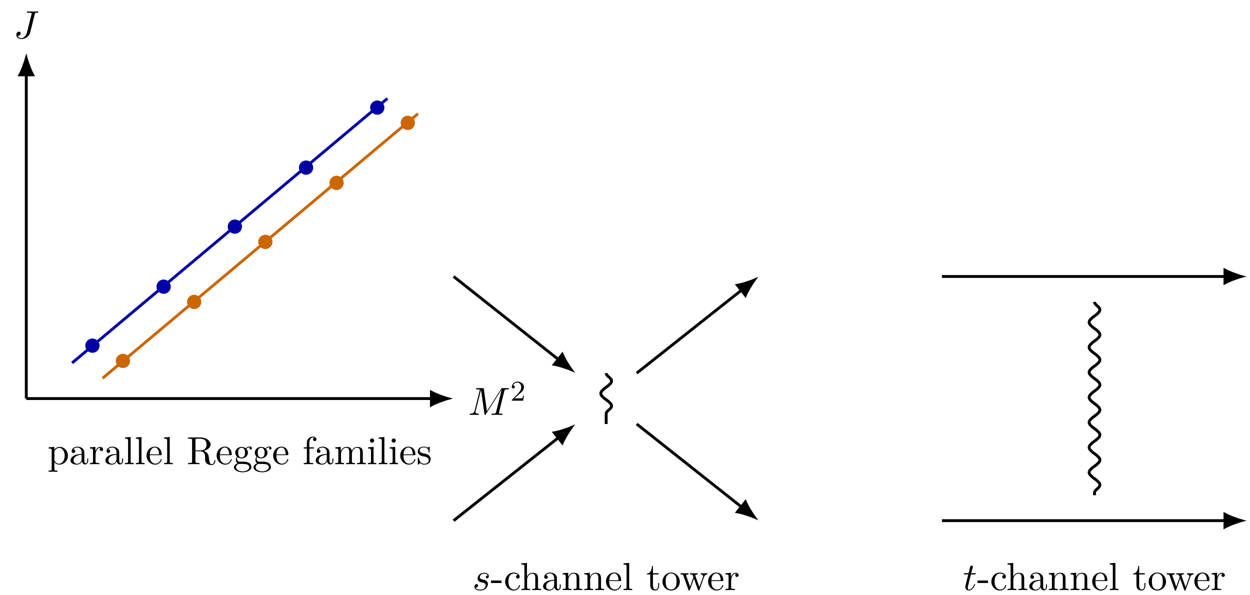

Figure 1.2: Parallel Regge families and the two channel expansions involved in hadronic duality.

1.3 One world sheet, two slicings

The natural response was to seek one geometrical history that could be read in either way. Represent a pion by a quark–antiquark pair and let time run upward. Instead of drawing only one-dimensional particle lines, fill in the region bounded by the quark trajectories. A horizontal slice through this two-dimensional history may show two incoming strings joining and later separating. A different slicing displays a string exchanged from one side to the other. The two Feynman-channel pictures are no longer independent mechanisms; they are different decompositions of one surface.

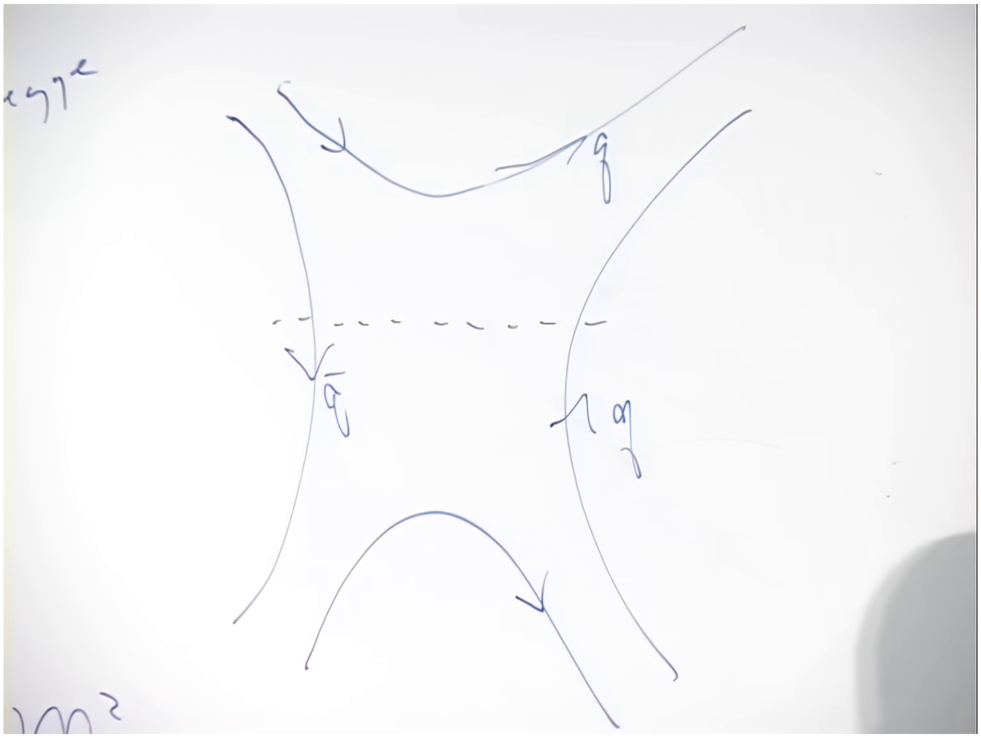

Figure 1.3: The quark-line topology on the board. Different cuts through the same two-dimensional history produce the resonance and exchange readings.

Lecture frame: 00:17:50.

The region between a quark and an antiquark supplies one further idea. At a fixed time, the slice contains two endpoints and something filling the whole interval between them. That object is one-dimensional: a string. During an interaction the string sweeps out the two-dimensional surface.

Definition 1.1. The spacetime history of a point particle is a *world line*. The spacetime history of a string is a *world sheet*. A string at one instant is a spatial slice of its world sheet.

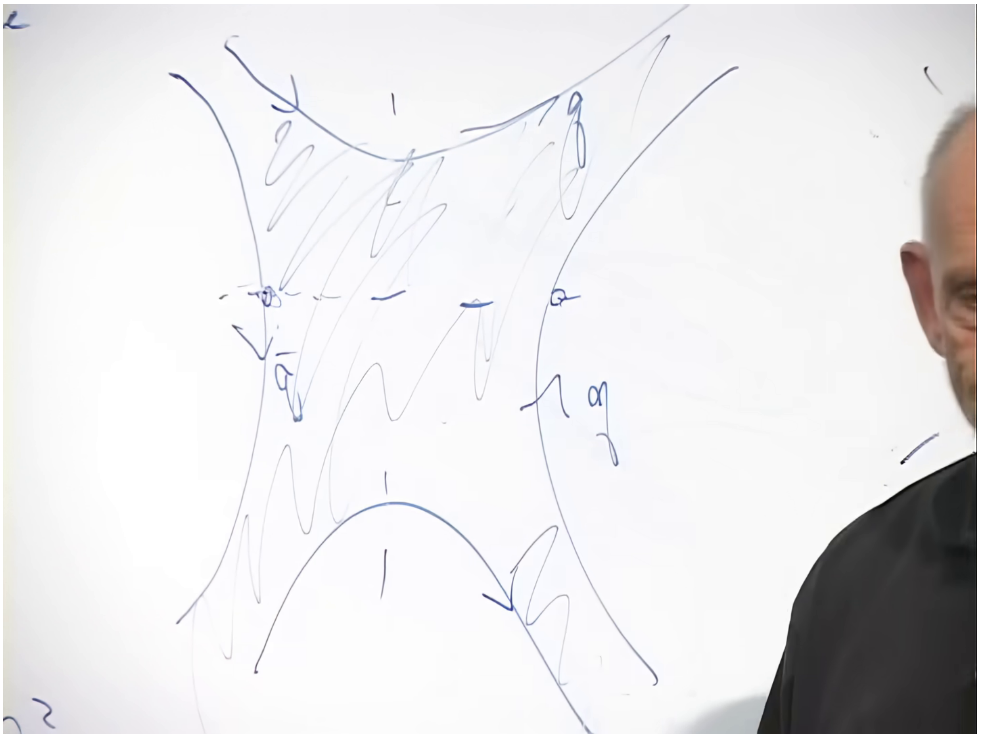

Figure 1.4: The board surface filled in between the endpoint trajectories. A time slice intersects it in a one-dimensional string.

Lecture frame: 00:19:15.

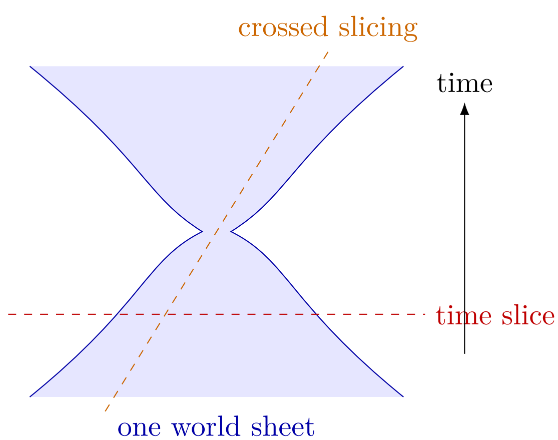

Figure 1.5: A clean world-sheet schematic. Distinct slicings organize the same surface into different intermediate-channel descriptions.

This geometric reconstruction was exploratory, not a deduction of the full theory from first principles. Different people reached the early string picture in different ways. Its strength was that the same extended object could address both observations: a string can rotate and deform, and its world sheet can encode the dual scattering channels.

Question from the audience. Why introduce a string-like bridge rather than an ordinary particle exchanged between the quarks?[a]

Response. At one instant there is not merely a localized exchanged particle. There is a quark, an antiquark, and structure at all points between them. That structure might be fundamentally continuous or might be a dense collection of constituents approximating a continuum; the first picture did not decide. What mattered was that the whole interval was occupied and that the observed trajectories showed no tendency to terminate like an ordinary shoelace with weights on its ends.

[a]Lecture timestamp: 00:21:20.

Question from the audience. Can the Standard Model describe pion scattering? [a]

Response. Yes, although usefulness depends on the energy regime. At low energy the Standard Model, through QCD and its low-energy descriptions, accounts for a great deal of pion scattering. At higher energies, where whole towers of hadronic exchanges matter, direct calculations become more difficult and a string-like organization can be more useful.

[a]Lecture timestamp: 00:22:45.

Question from the audience. Did the early experiments establish that a Regge trajectory never ends?[a]

Response. No. They established only that the observed sequence did not yet show an endpoint. The inference of indefinite stretchability was a suggestive extrapolation, not a measured infinity.

[a]Lecture timestamp: 00:23:37.

1.4 Flux tubes and the change of scale

Modern QCD gives the old picture a field-theoretic interpretation. A quark and an antiquark act as opposite sources of the gluon field. In ordinary electrodynamics, electric flux spreads sideways as opposite charges are separated. The field weakens with distance, so the region between the charges does not retain a fixed transverse profile. QCD is nonlinear. Its dynamics can confine the relevant chromoelectric flux to a narrow tube joining the quark and antiquark.

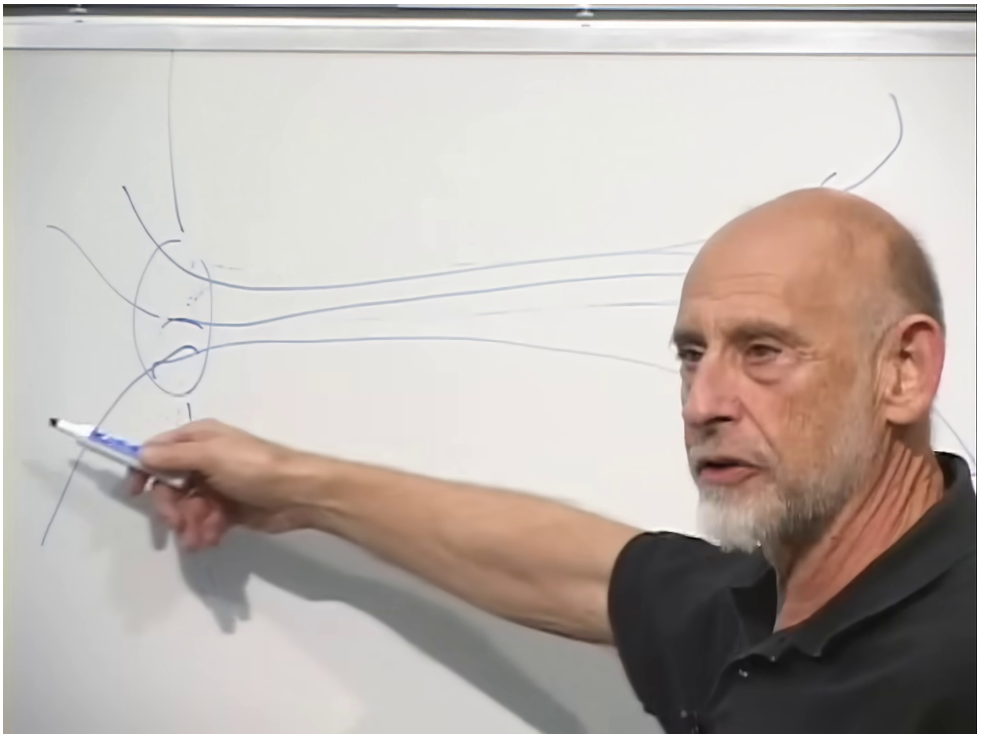

Figure 1.6: Quark and antiquark field lines drawn as a narrow flux tube rather than a freely spreading Coulomb field.

Lecture frame: 00:25:15.

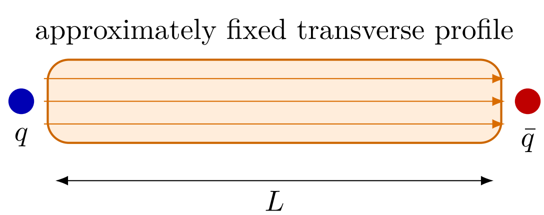

Figure 1.7: A confined flux tube lengthens without spreading. Fixed energy per unit length gives $E = TL$.

The particle language and field language say the same thing differently. In the field description, the tube retains roughly the same field strength and cross-sectional area as its endpoints separate. In a constituent picture, the stretching energy creates more gluonic structure between the endpoints. An ordinary rubber band has a fixed number of molecules, so its molecules separate until the material breaks. Imagine instead that a new molecule were inserted whenever a gap opened. The density along the band could remain fixed while the band became longer. That is the intended analogy for the gluon field: the tube gains length rather than thinning toward rupture.

Question from the audience. Would a string that made up an electron also be composed of gluons?[a]

Response. No such conclusion follows. The hadronic string is a QCD object on the scale of a proton. A hypothetical fundamental string relevant to quantum gravity would live near the vastly smaller Planck scale, and there is no evidence here that an electron is a hadronic gluon string. The striking fact is that related mathematics arose at both scales, not that the physical objects are identical.

[a]Lecture timestamp: 00:28:08.

Attempts to turn the hadronic picture into a complete interacting string theory were promising but did not quite produce the desired theory of hadrons. The mathematical model repeatedly contained a massless particle of spin two. In a hadronic theory that state looked like an unwanted prediction; as a theory of gravity it had exactly the quantum numbers of the graviton. Joel Scherk and John Schwarz made the decisive reinterpretation: perhaps the theory was telling us about quantum gravity rather than hadrons. The hadronic and fundamental applications are physically different, even though the early hadron problem supplied the route to the mathematics.

Question from the audience. How do the flux lines go from a positive endpoint to a negative endpoint?[a]

Response. They leave one source and enter the other, just as magnetic field lines run from the north pole to the south pole of a bar magnet. The opposite endpoint labels express the orientation of the flux; a bar magnet likewise has two opposite poles rather than two north poles.

[a]Lecture timestamp: 00:31:46.

Question from the audience. Is the string continuous or discrete? [a]

Response. Quantum theory makes that distinction subtle. The electromagnetic field is represented as a continuous field and also has discrete quanta. In the same way, a string may be treated continuously while its excitations are quantized. Both descriptions can be valid, depending on the question being asked.

[a]Lecture timestamp: 00:32:45.

Question from the audience. If the angle of the spacetime slice through a world sheet is changed, is the intersection still a string?[a]

Response. Yes. Changing the simultaneity slice is a Lorentz transformation. The new slice intersects the surface in a string seen in a different state of motion. Even a string whose center of mass is at rest is not internally motionless; it may vibrate. "At rest" refers only to its center of mass.

[a]Lecture timestamp: 00:34:20.

1.5 Two expansions of relativistic energy

A direct relativistic description of an extended, vibrating object is awkward. The useful strategy is to find a frame in which its relevant motion has the form of ordinary nonrelativistic mechanics. Begin with a single particle. Nonrelativistically one writes

$$E_{\mathrm{nr}} = \frac{\boldsymbol{p}^2}{2m} + B, \tag{1.5}$$

where B is an internal or binding energy independent of the overall momentum. For several noninteracting particles, both the kinetic terms and the internal constants are added. Since only energy differences govern the motion, a state-independent additive constant can often be suppressed.

The exact relativistic energy is

$$E = \sqrt{\boldsymbol{p}^2c^2 + m^2c^4}. \tag{1.6}$$

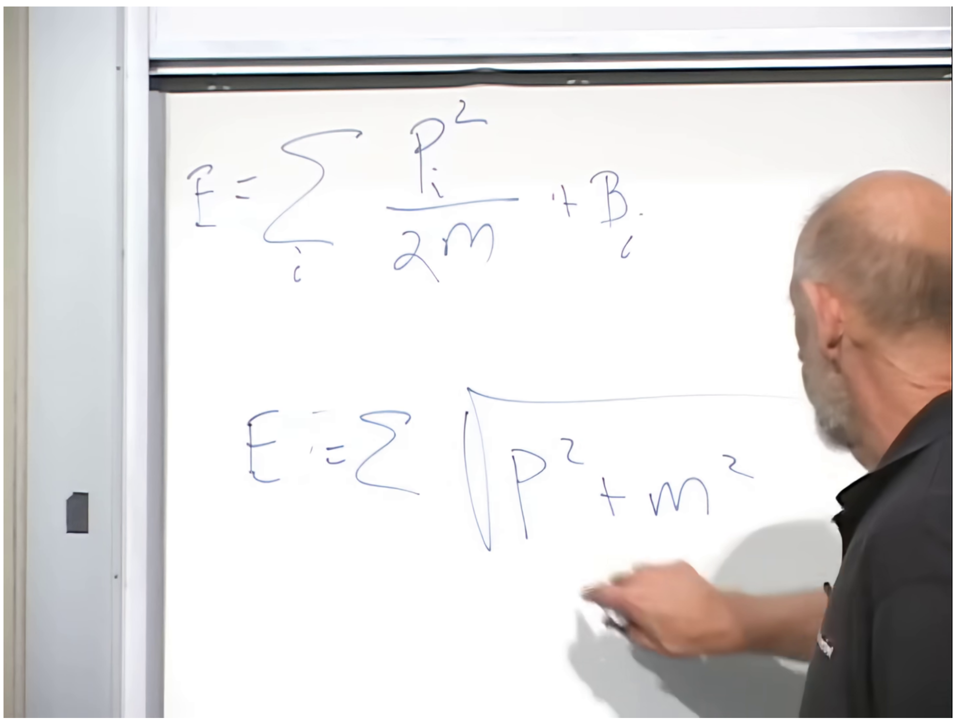

Figure 1.8: The nonrelativistic kinetic energy and the exact relativistic square root, with the powers of c corrected during the discussion.

Lecture frame: 00:38:45.

Question from the audience. Should the relativistic formula contain p^2c^2 and m^2c^4, rather than the powers initially written on the board? [a]

Response. Yes. Dimensional consistency fixes the expression to $E^2 = p^2c^2 + m^2c^4$. After recovering the nonrelativistic limit we shall set $c = 1$, but keeping the powers here makes the expansion transparent.

[a] Lecture timestamp: 00:39:12.

When $p^2c^2 \ll m^2c^4$, factor out m^2c^4 and use $\sqrt{1+\epsilon} \simeq 1 + \epsilon/2$:

$$
\begin{aligned}
E &= mc^2\sqrt{1+\frac{p^2}{m^2c^2}} \\
&\simeq mc^2 + \frac{p^2}{2m}.
\end{aligned}
\tag{1.7}
$$

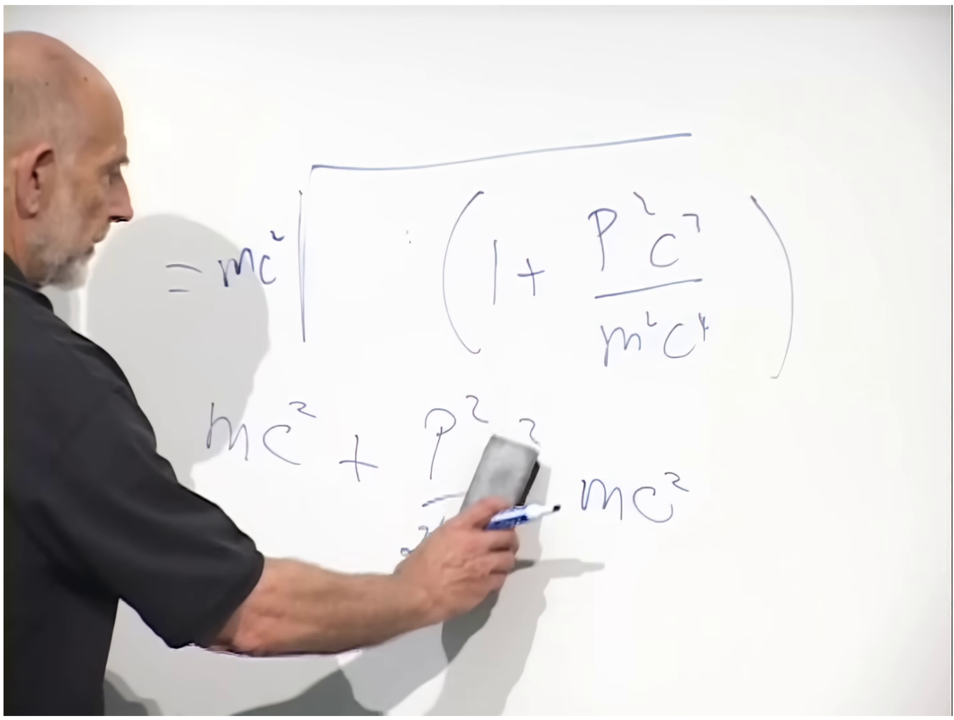

Figure 1.9: The small-momentum expansion recovering rest energy plus the familiar nonrelativistic kinetic term.

Lecture frame: 00:41:30.

This approximation requires more than a slowly moving center of mass. A box may drift slowly while particles inside it move near the speed of light. In that case the internal dynamics are still relativistic. The ordinary nonrelativistic limit requires all relevant constituent velocities to be small in the chosen frame.

There is another useful limit. Boost the entire system along the z-axis until every constituent has a large positive longitudinal momentum P_{zi}. Any constituent initially moving backward is carried forward by a sufficiently large boost. The transverse momenta and rest masses remain fixed:

$$
P_{zi} \gg |p_{xi}|, |p_{yi}|, m_i, \qquad p_{\perp i}^2 = p_{xi}^2 + p_{yi}^2, \tag{1.8}
$$

where from this point onward $c = 1$. Expanding for large P_{zi} gives

$$
\begin{aligned}
E_i &= \sqrt{P_{zi}^2 + p_{\perp i}^2 + m_i^2} \\
&= P_{zi}\sqrt{1 + \frac{p_{\perp i}^2 + m_i^2}{P_{zi}^2}} \\
&\simeq P_{zi} + \frac{p_{\perp i}^2 + m_i^2}{2P_{zi}}.
\end{aligned}
\tag{1.9}
$$

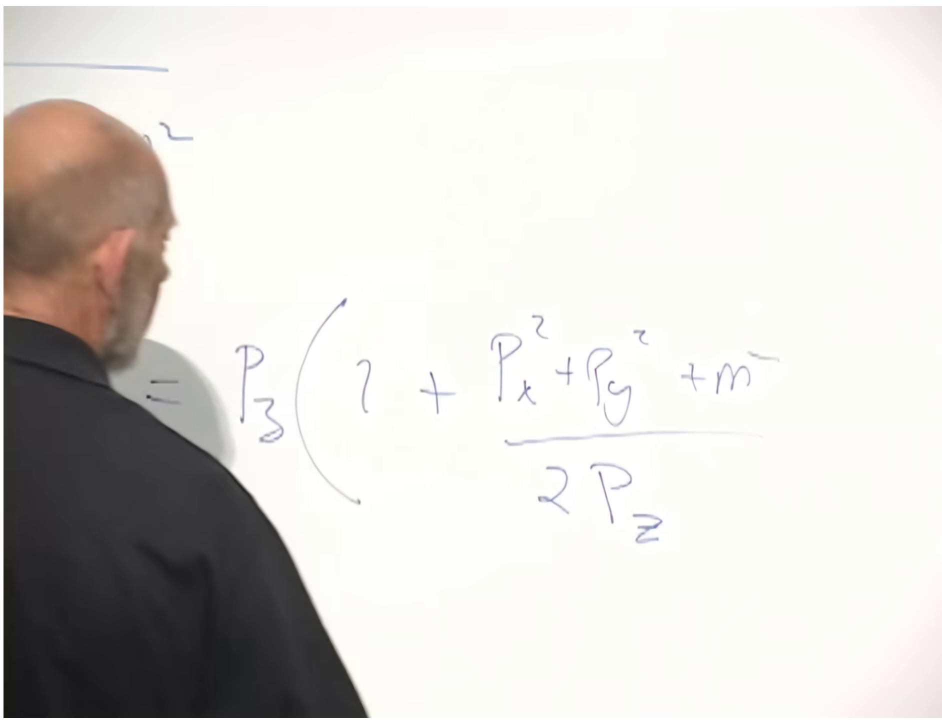

Figure 1.10: The large-P_z square-root expansion, with transverse momentum and mass retained as the small quantities.

Lecture frame: 00:46:45.

For many constituents,

$$
E \simeq \sum_i P_{zi} + \sum_i \frac{p_{\perp i}^2}{2P_{zi}} + \sum_i \frac{m_i^2}{2P_{zi}}. \tag{1.10}
$$

The first term is the conserved total longitudinal momentum. It is enormous, but it is also a fixed additive contribution. Keeping or subtracting it changes no energy difference, just as adding any other conserved number to a Hamiltonian would not change the equations of motion. The dynamical part is therefore

$$
\begin{aligned}
H_\perp = E - P_z &\simeq \sum_i \left(\frac{p_{\perp i}^2}{2P_{zi}} + \frac{m_i^2}{2P_{zi}} \right), \\
P_z &= \sum_i P_{zi}.
\end{aligned}
\tag{1.11}
$$

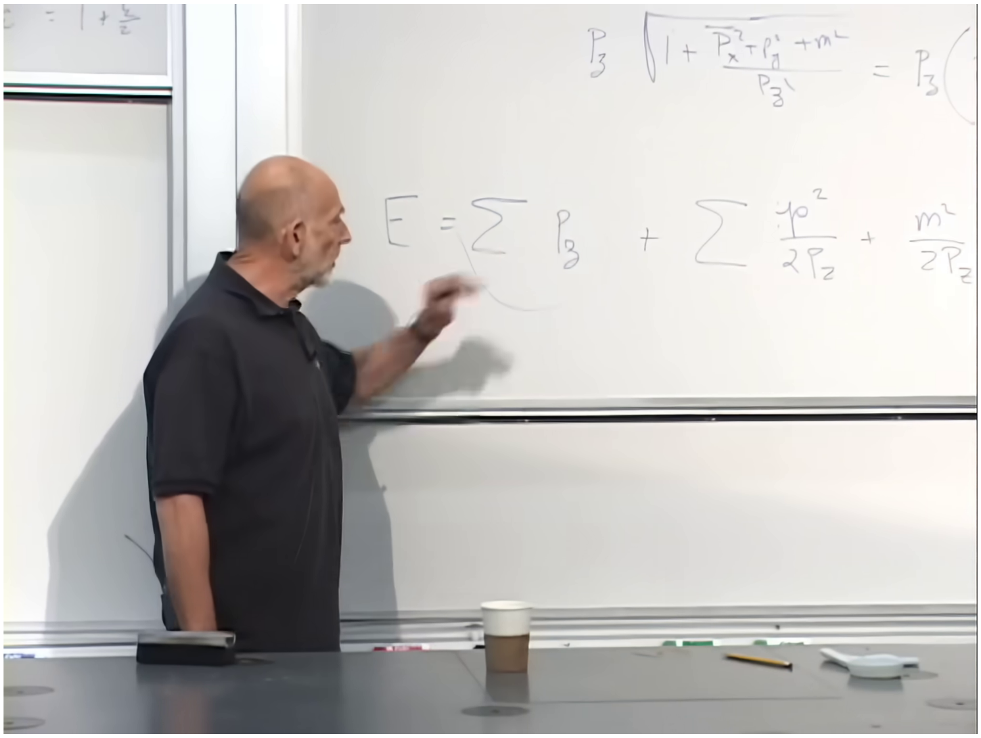

Figure 1.11: The many-particle decomposition into conserved longitudinal momentum, transverse kinetic energy, and mass-dependent internal terms.

Lecture frame: 00:49:16.

1.6 The infinite-momentum frame

The preceding limit was historically called the infinite-momentum frame and is now usually formulated in light-cone coordinates. Its meaning can be read directly from the Hamiltonian. In quantum mechanics,

$$i\hbar\frac{\partial}{\partial t}\left|\psi\right\rangle = H\left|\psi\right\rangle . \tag{1.12}$$

All nontrivial terms in $H_\perp$ scale as $1/P_z$. As the boost grows, energy splittings shrink and changes occur more slowly. This is time dilation. A radioactive system appears to decay more slowly when highly boosted; one can compensate by observing it for a proportionally longer time. The slowing does not remove the dynamics. It rescales their clock.

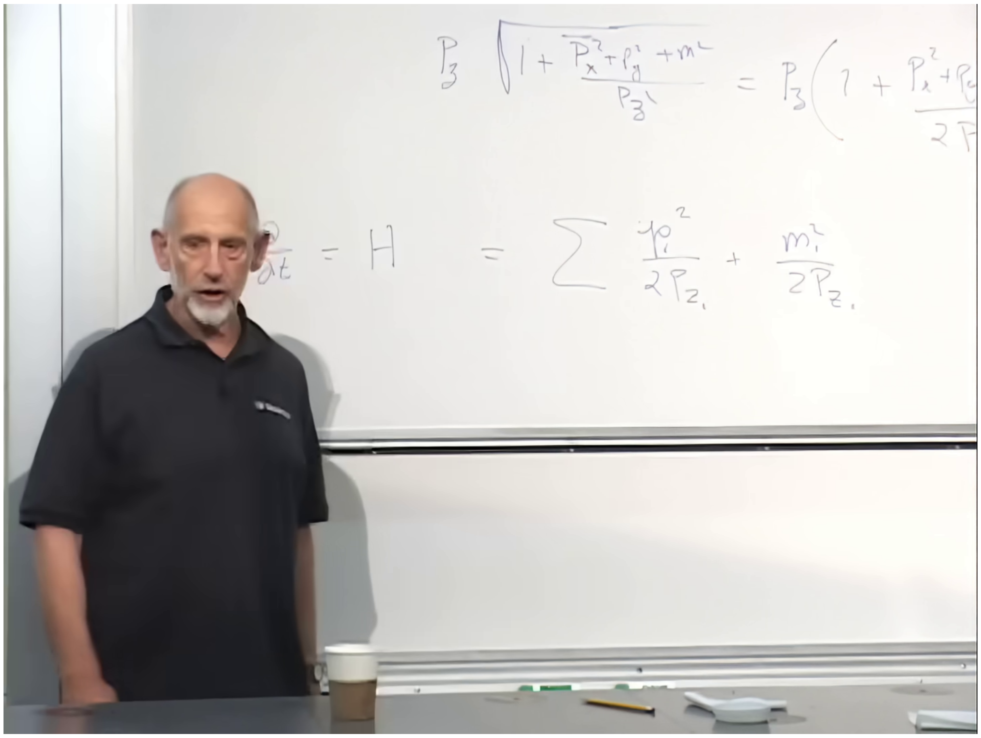

Figure 1.12: The board identifies the reduced energy with the Hamiltonian that generates the slowed time evolution of the boosted system.

Lecture frame: 00:52:57.

The transverse term has precisely the nonrelativistic form $p^2/(2m)$, but the role of mass is played by P_z. This is physically sensible: a larger longitudinal momentum makes a particle harder to deflect sideways, so it acts as transverse inertia. The term $m_i^2/(2P_{zi})$ does not depend on the two-dimensional state of motion and therefore plays the role of an internal or binding energy. Relativistic motion in three spatial dimensions has become nonrelativistic-looking motion in the two dimensions perpendicular to the boost.

This is the permission needed to use an elementary masses-and-springs model for a relativistic string. The model is not pretending that the string is slow in its rest frame. It uses the frame in which the transverse dynamics have the required nonrelativistic structure.

1.7 An open string made from beads and springs

Take an open string, meaning a string with two endpoints. Something may later be attached to those endpoints, but the present concern is the string itself. Approximate it by beads at transverse positions $\boldsymbol{X}_i(t) = (X_i(t), Y_i(t))$, joined by identical springs. A discrete Hamiltonian is

$$H_N = \sum_{i=1}^{N} \left[\frac{\mu_N}{2} \dot{\boldsymbol{X}}_i^2 + \frac{k_N}{2} \left(\boldsymbol{X}_{i+1} - \boldsymbol{X}_i \right)^2 \right]. \tag{1.13}$$

The kinetic term alone would let the beads fly apart. The nearest-neighbor Hooke term is what makes them a string.

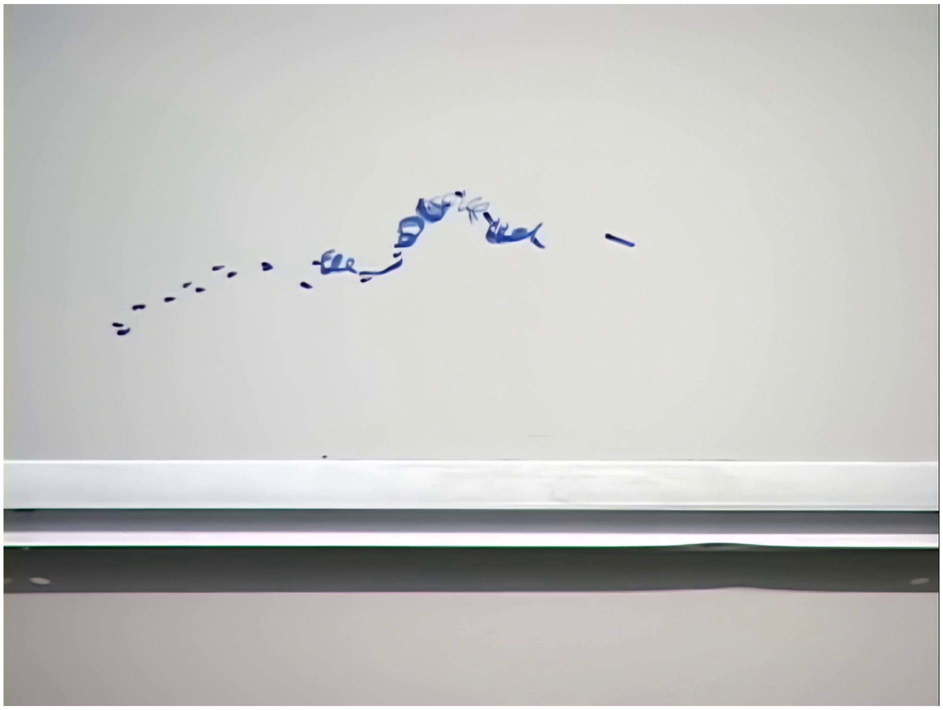

Figure 1.13: The first bead-and-spring sketch for an open string with two transverse coordinates.
Lecture frame: 01:00:30.

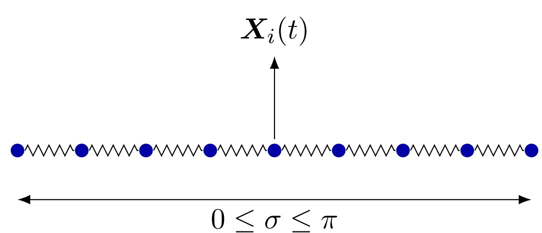

Figure 1.14: Discrete regulator for the open string. The bead mass and spring constant scale with the spacing before the continuum limit is taken.

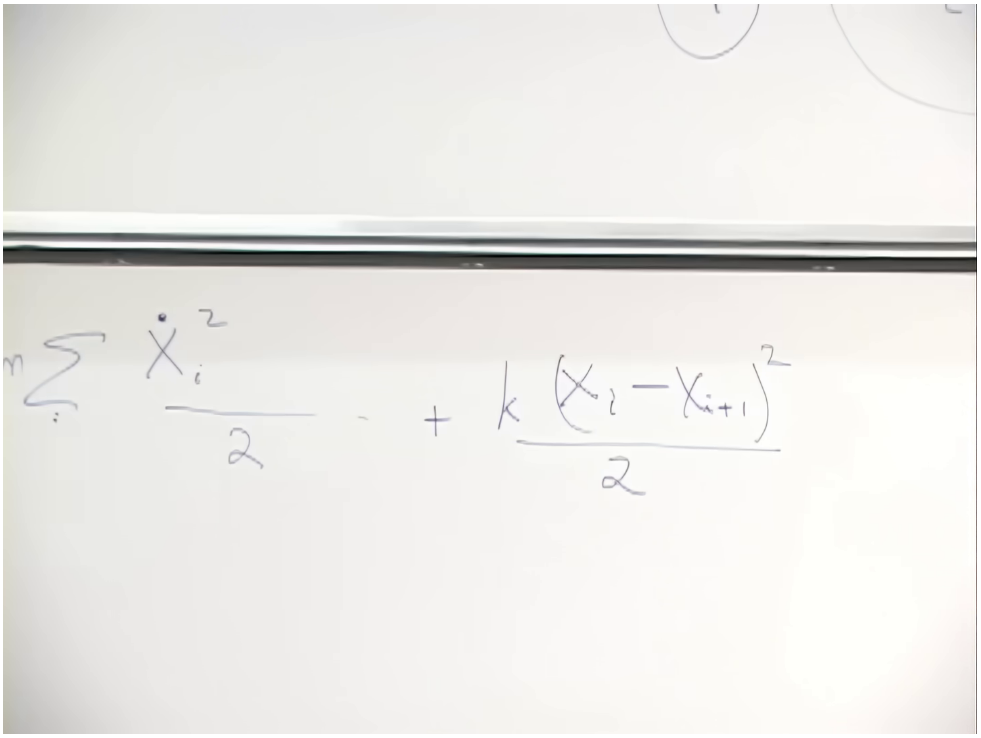

Figure 1.15: The discrete kinetic and Hooke terms written together before the continuum limit.

Lecture frame: 01:01:30.

As the number of beads increases, each bead mass μ_N must tend to zero so that the whole string retains finite inertia. At the same time the spring constant k_N must grow. A long strip of rubber is easier to stretch by a fixed amount than a tiny segment of the same material; neighboring points that are brought closer together must therefore be connected more stiffly if the continuum material is to remain the same.

Introduce a coordinate σ along the string. Its numerical range is a convention. Choosing

$$0 \leq \sigma \leq \pi \tag{1.14}$$

is convenient because a closed string can later be parameterized from 0 to 2π. The sum becomes an integral, and the nearest-neighbor difference becomes a derivative:

$$\sum_i \longrightarrow \int_0^\pi d\sigma, \qquad \boldsymbol{X}_{i+1} - \boldsymbol{X}_i \longrightarrow \partial_\sigma \boldsymbol{X}. \tag{1.15}$$

After absorbing the spacing-dependent constants into the normalization of t, σ, and $\boldsymbol{X}$, the continuum Hamiltonian has the form

$$H_{\text{string}} = \frac{1}{2}\int_0^\pi d\sigma \left[(\partial_t \boldsymbol{X})^2 + (\partial_\sigma \boldsymbol{X})^2\right]. \tag{1.16}$$

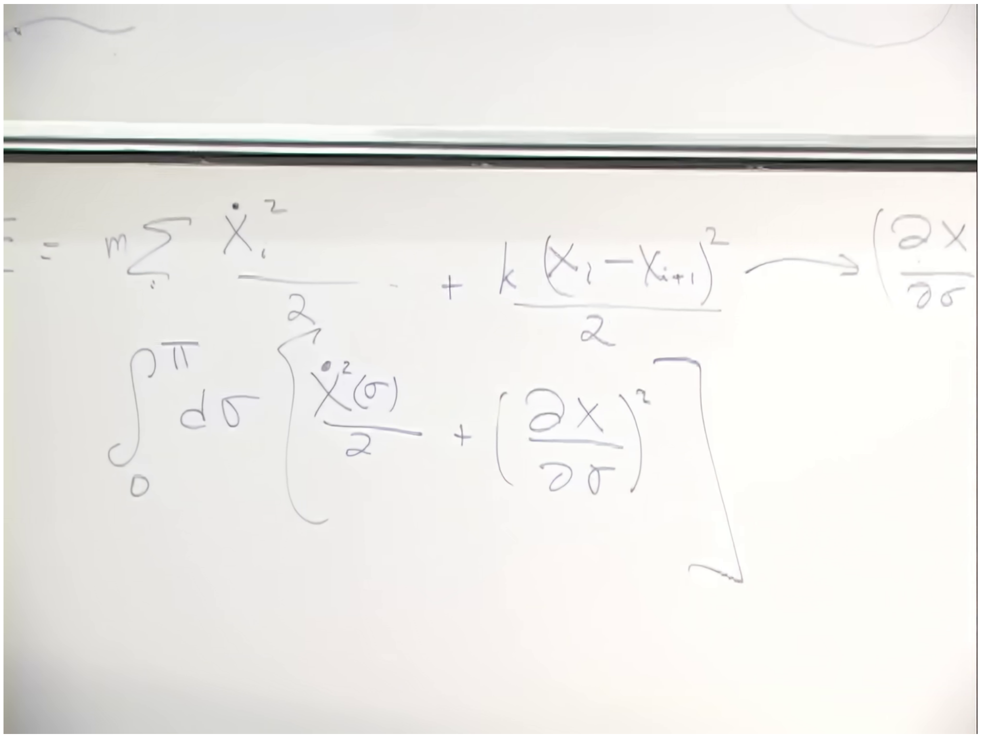

Figure 1.16: The board passage from the bead sum to $\int_0^\pi d\sigma$, with the nearest-neighbor difference replaced by $\partial_\sigma \boldsymbol{X}$.

Lecture frame: 01:05:00.

The first term is kinetic energy in the two transverse directions. The second measures stretching along the string. Both terms carry a plus sign because this is the Hamiltonian. In the Lagrangian the potential term would enter with a minus sign. The precise coefficients depend on conventions and were not fixed at this stage; the kinetic-plus-stretching structure is what will matter.

Separate the center-of-mass motion from the relative motion. Translating every point by the same amount moves the particle as a whole. Oscillations, bending, and relative stretching are internal motions. A proton, an atom, or any other composite object can be called a particle even though its constituents move internally. The string is no different in that respect.

1.8 Why the Hamiltonian measures mass squared

For a slowly moving object, rest energy is proportional to its mass. In the infinite-momentum description, however, the internal contribution appears as $M^2/(2P_z)$. Once the common longitudinal momentum and its associated time rescaling have been fixed, the internal string Hamiltonian is proportional to the invariant mass squared of the whole object:

$$H_{\text{int}} \propto M^2. \tag{1.17}$$

[2]

[2]Editorial clarification: More explicitly, light-cone kinematics gives an invariant relation of the form $M^2 =$

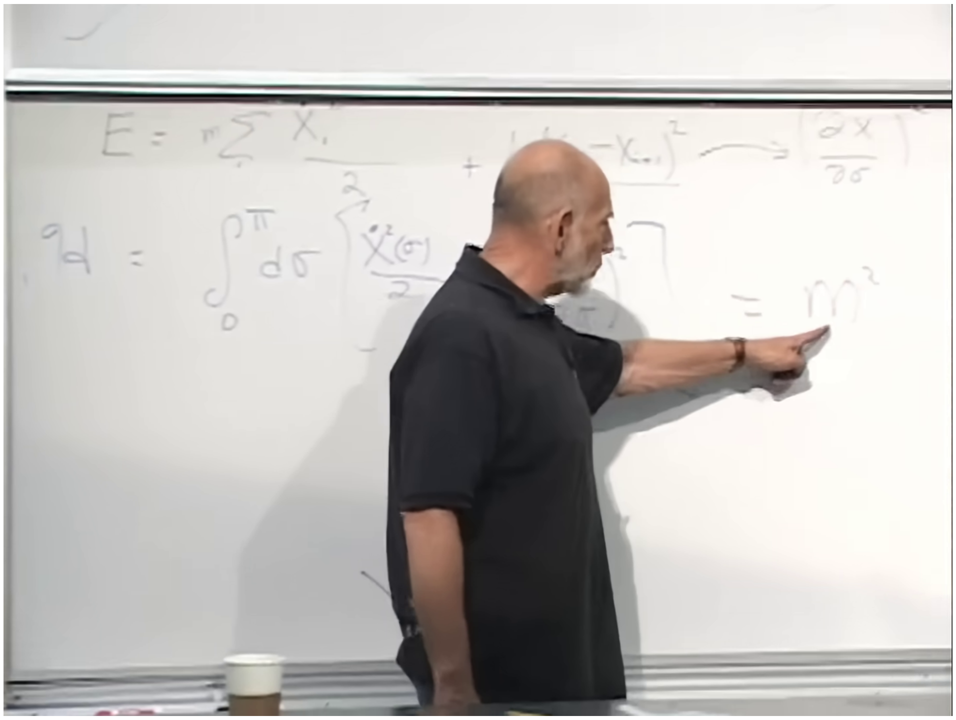

Figure 1.17: The board identifies the internal string energy with the mass squared of the complete string state.

Lecture frame: 01:10:30.

A vibrating string is a collection of coupled oscillators. Its complete quantization is postponed, but the important spectral fact is already visible. Harmonic-oscillator energies rise in discrete steps. Since those internal energies contribute to M^2, string excitations naturally produce discrete steps in mass squared. This is the structure that the Regge plots announced at the beginning.

Question from the audience. Is M the mass of the whole string, and would doubling that mass quadruple the corresponding M^2 contribution? [a]

Response. Yes. M is the rest mass of the complete string state, including vibration, stretching, and every other internal contribution. The effective boosted Hamiltonian is proportional to M^2, so a state with twice the mass has four times that mass-squared value.

[a] Lecture timestamp: 01:12:10.

Question from the audience. How can this description include a photon, whose rest mass is zero?[a]

Response. A massless state corresponds to $M = 0$. How the positive oscillator contributions can combine to produce such a state is a significant question for the quantized theory, not something settled by the classical model here. At this point M simply denotes the total rest mass, and quantum mechanics has not yet been introduced.

[a] Lecture timestamp: 01:12:52.

$2P^+P^- - \boldsymbol{P}_\perp^2$. In the center-of-mass transverse frame and at fixed P^+, the light-cone Hamiltonian P^- is therefore proportional to M^2. The lecture suppresses these normalization details and keeps the proportionality needed for the argument.

Now stretch a nonmoving string uniformly to physical length L. The kinetic term vanishes, and over the parameter interval of length π,

$$|\partial_\sigma \boldsymbol{X}| \simeq \frac{L}{\pi}. \tag{1.18}$$

Consequently,

$$\begin{aligned} H_{\text{stretch}} &\propto \int_0^\pi d\sigma\, (\partial_\sigma \boldsymbol{X})^2 \\ &\simeq \pi \left(\frac{L}{\pi}\right)^2 \propto L^2. \end{aligned} \tag{1.19}$$

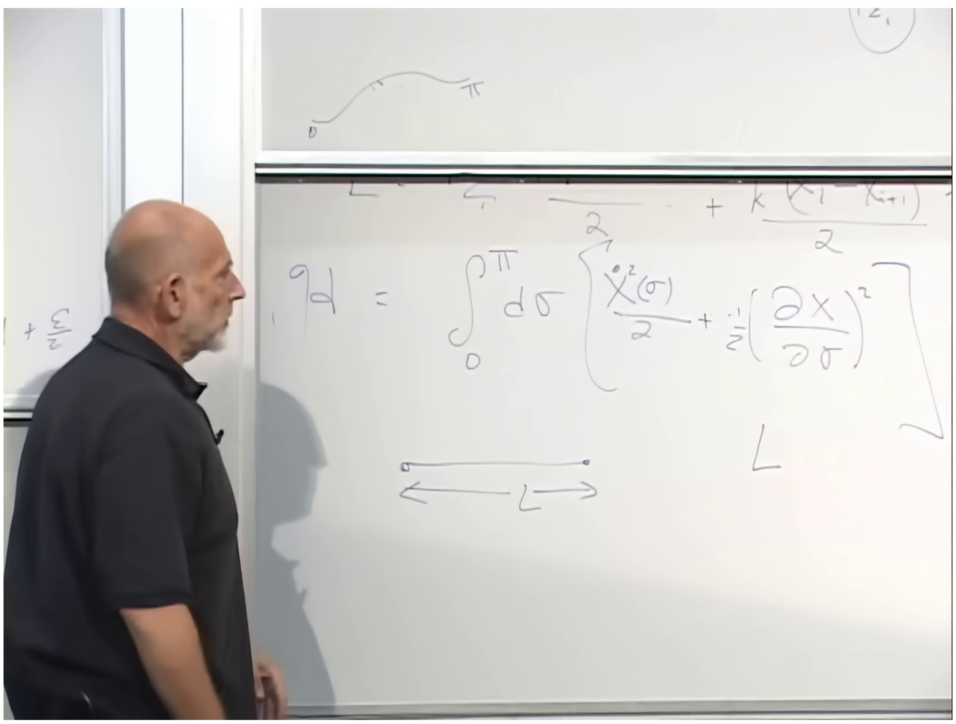

Figure 1.18: The uniformly stretched configuration used to infer $H_{\text{stretch}} \propto L^2$ and hence $M \propto L$. *Lecture frame: 01:15:30.*

This is Hooke-law behavior in the effective transverse mechanics. But the quantity on the left is proportional to M^2, not M. It follows that

$$M^2 \propto L^2, \qquad M \propto L. \tag{1.20}$$

In the rest frame the energy is $E_{\text{rest}} = M$, in units $c = 1$. Writing the constant of proportionality as the string tension T,

$$E_{\text{rest}} = TL. \tag{1.21}$$

The result matches the flux-tube picture. A uniform electric or magnetic field carries a uniform energy density. If its flux is confined to a tube of fixed cross section, each additional unit of length adds the same amount of energy. Equivalently, a constituent picture in which new degrees of freedom fill each newly opened gap has fixed energy per unit length. String tension is that energy per unit length.

This conclusion is indirect. It does not claim that an arbitrary rubber band driven to relativistic speeds will automatically acquire $E = TL$. The argument begins with the transverse nonrelativistic description that follows from the large boost, identifies its internal energy with M^2, and only then translates the result back to rest-frame energy.

1.9 The same object in two frames

The distinction between M and M^2 is easy to lose, so it is worth placing the two expansions side by side. With $c = 1$,

$$E = \sqrt{p^2 + M^2} \simeq M + \frac{p^2}{2M}, \qquad p \ll M, \tag{1.22}$$

$$E = \sqrt{P_z^2 + M^2} \simeq P_z + \frac{M^2}{2P_z}, \qquad P_z \gg M. \tag{1.23}$$

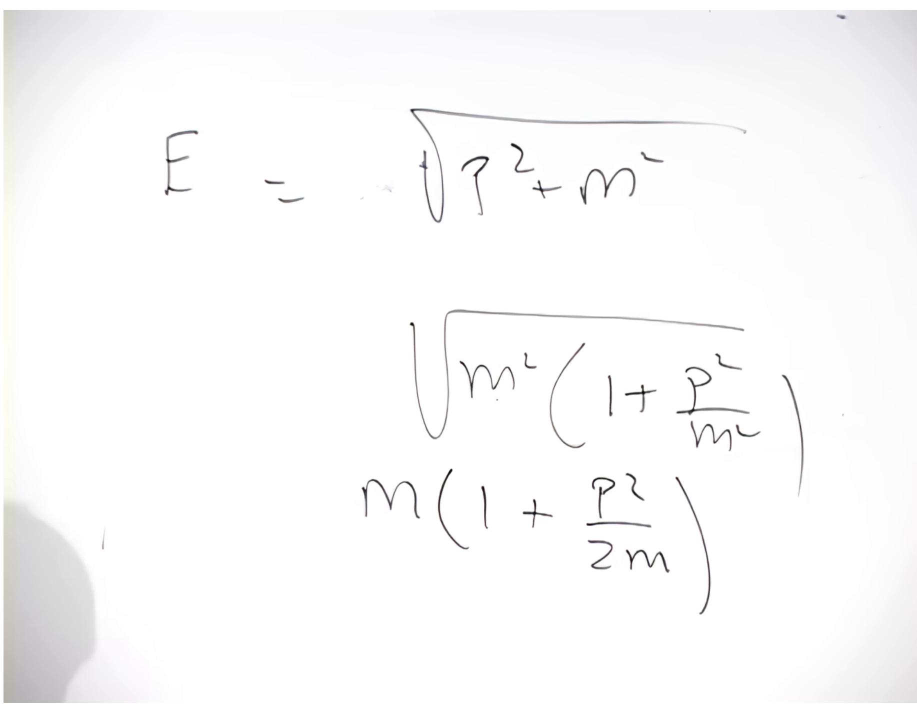

Figure 1.19: The small-momentum expansion repeated after the break, where internal rest energy is read as M.

Lecture frame: 01:28:00.

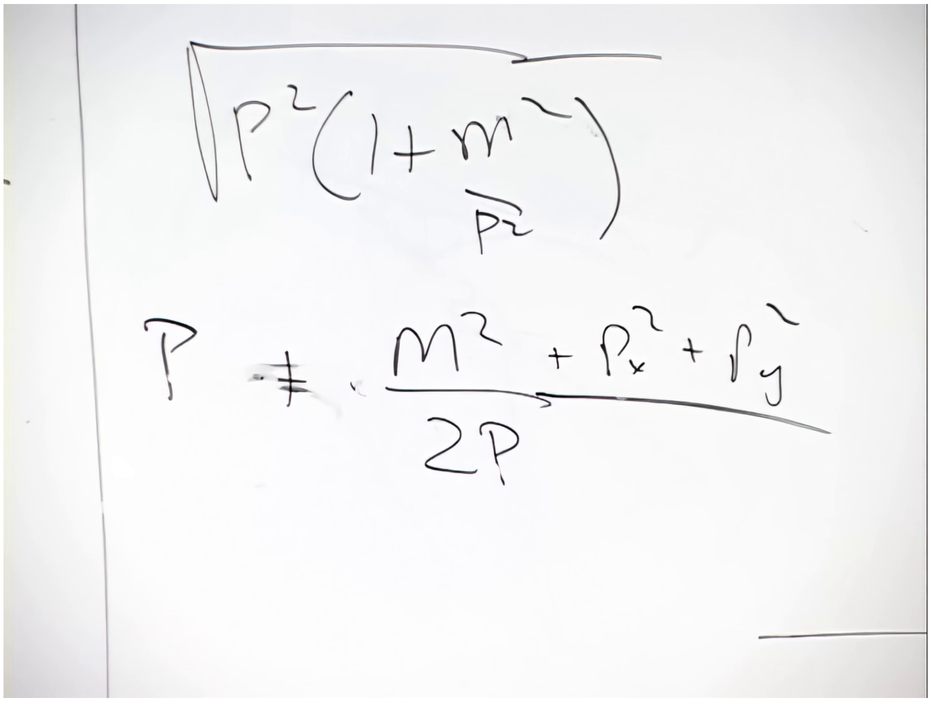

Figure 1.20: The complementary large-momentum expansion, where the residual energy is proportional to $M^2/(2P_z)$.

Lecture frame: 01:31:00.

The first expansion identifies rest energy with mass and ordinary inertia with M. In the second, the conserved P_z term is removed, transverse inertia is P_z, and internal energy is represented by $M^2/(2P_z)$. The mass remains proportional to the string length in either frame. What changes is which function of the mass appears as the residual Hamiltonian.

Question from the audience. A Lorentz boost should not change the physics. Why does the boosted string appear to obey a different energy law? [a]

Response. It does not obey different physics. In both descriptions $M \propto L$ and $M^2 \propto L^2$. In the rest frame the relevant energy is M; after a large boost the conserved P_z is separated off and the remaining Hamiltonian is proportional to M^2. The boost changes the useful description, not the invariant mass–length relation.

[a] Lecture timestamp: 01:25:35.

1.10 A superconducting model of confinement

A superconductor expels magnetic field from its bulk. If magnetic flux is trapped through a sufficiently narrow channel before the material becomes superconducting, the flux cannot spread sideways and remains confined to a tube. A useful Gedanken experiment is to drill a narrow hole through a normal-state sample, insert a long bar magnet through it, cool the sample, and then withdraw the magnet. The construction is schematic rather than a practical laboratory recipe. Its purpose is to isolate a flux line whose field strength and energy density are approximately uniform along its length.

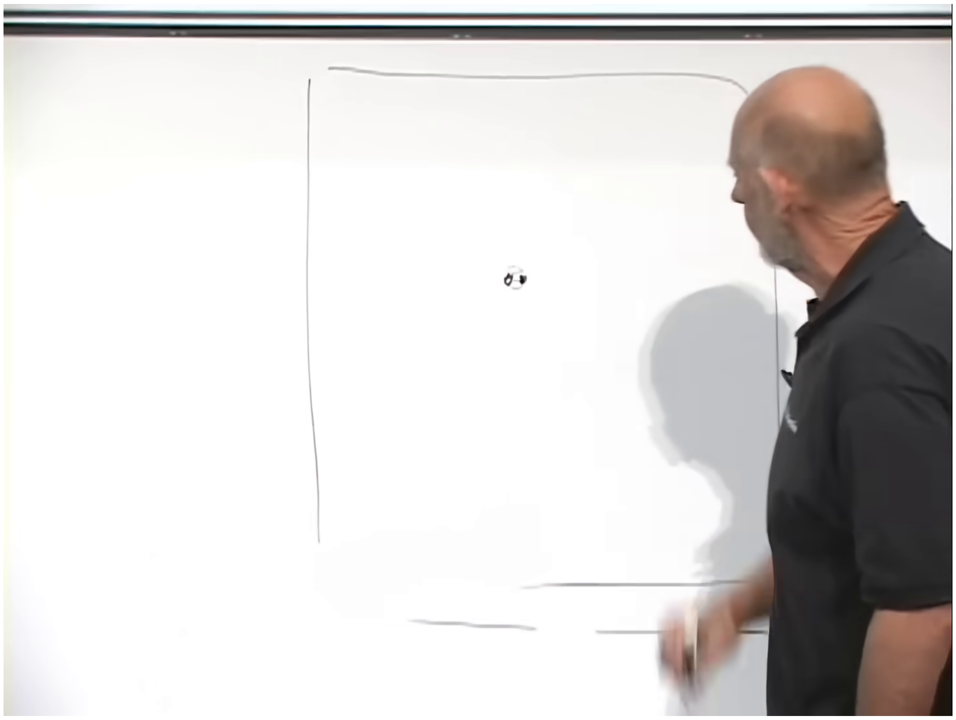

Figure 1.21: The superconducting flux-line construction and the corresponding monopole–antimonopole confinement picture.

Lecture frame: 01:35:30.

Suppose, as a further thought experiment, that a magnetic monopole and antimonopole could be placed in the superconducting medium and separated. The magnetic flux between them would be squeezed into a tube. Since each unit of tube length carries the same energy,

$$V(R) = T_{\text{flux}} R. \tag{1.24}$$

The force $F = -dV/dR$ has constant magnitude. Pulling the pair apart becomes increasingly expensive, so the pair is confined. This is the behavior sought for quarks in a hadron.

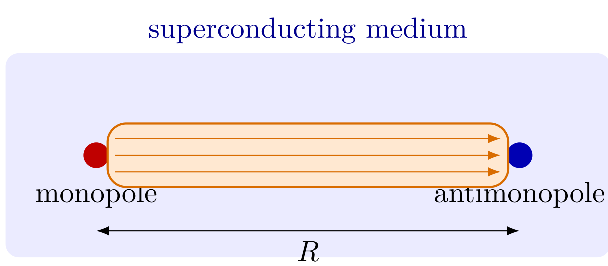

Figure 1.22: Flux confinement gives a uniform tube and a linear potential $V(R) = T_{\text{flux}} R$.

Question from the audience. What plays the role of the superconducting medium in the hadron?[a]

Response. The proposed analogy reverses the electric and magnetic roles. An ordinary superconducting condensate is electrically charged and confines magnetic flux. In the dual-superconductor picture of QCD, a condensate of chromomagnetic objects confines the chromoelectric charge of quarks. Such objects do not appear as isolated particles because they are represented as condensed in the vacuum.

[a]Lecture timestamp: 01:36:57.

[3]

Question from the audience. Is energy proportional to length a general rule for any ordinary string made highly relativistic?[a]

Response. No. One would first need a consistent relativistic theory of that material, and its behavior would depend on the details. Linear energy is special to a configuration whose density stays fixed as it is extended, or to flux prevented from spreading transversely. Pulling it apart creates more constituents or more uniform field-filled length rather than merely increasing the separation of a fixed set of molecules.

[a]Lecture timestamp: 01:38:10.

1.11 Transverse degrees of freedom and the size of a particle

The light-cone construction writes the string directly in terms of its transverse coordinates. There is no separate, freely specifiable longitudinal oscillator analogous to each transverse oscillator. This reduction by one direct spatial dimension is one of the remarkable features of the formalism; the complete theory nevertheless has to satisfy Lorentz invariance.

Question from the audience. What happens to residual motion along the z-direction? [a]

Response. In the light-cone description, longitudinal motion is not introduced as an independent degree of freedom. The physical oscillations are described transversely, while constraints determine the longitudinal information. This economy is related in spirit to the idea that gravitating systems may admit a description with one fewer direct dimension, although that connection is not derived here.

[a]Lecture timestamp: 01:40:10.

Question from the audience. How many points are actually known on the Regge plots? [a]

Response. The exact maximum depends on the family; a rough number is about seven. The sequence is not experimentally endless. The relevant fact is that the known points do not display the kind of visible failure or breakup endpoint familiar from ordinary macroscopic material.

[a]Lecture timestamp: 01:41:15.

Hadronic phenomenology supplied the historical motivation, but the mathematical string theory is not required to reproduce hadrons in every detail. Once the formalism is built, one asks what it

[3]Editorial clarification: The dual-superconductor description is an influential confinement mechanism and is presented here in the historical language of the lecture. It should be read as a physical picture of the QCD vacuum, not as a complete derivation of confinement from first principles.

actually describes. Its massless spin-two state and other properties point more naturally toward gravity.

The apparent contrast between hadrons and electrons can be quantified with the energy of a rigid rotation,

$$E_{\text{rot}} = \frac{J^2}{2I}, \qquad I \sim MR^2. \tag{1.25}$$

Angular momentum is quantized. Writing J^2 schematically as $\hbar^2 n^2$, the first rotational step is of order

$$\Delta E_{0\to 1} \sim \frac{\hbar^2}{2I}. \tag{1.26}$$

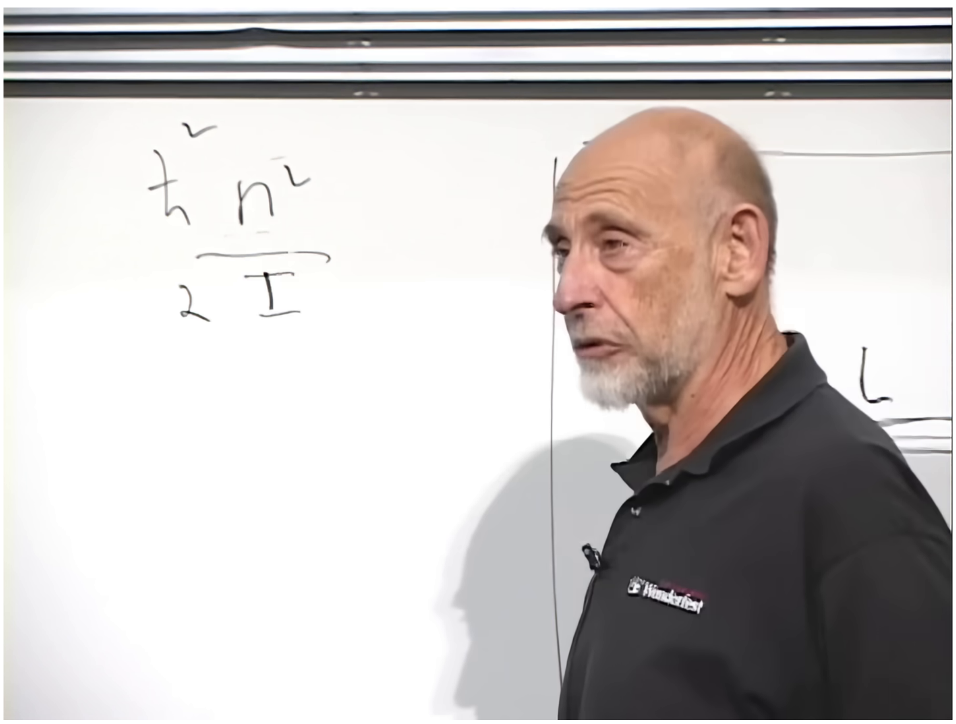

Figure 1.23: The moment-of-inertia estimate for the first rotational excitation, $\Delta E \sim \hbar^2/(2I)$.
Lecture frame: 01:44:00.

For fixed mass, a smaller radius means a much smaller moment of inertia and a much larger excitation energy. Hadrons are large on the scale of their internal dynamics, so a rotational excitation costs roughly a GeV and is accessible in particle collisions. If electrons possess any spatial substructure, experimental bounds make its scale far smaller, and rotational excitations would lie far beyond available energies. A basketball has the opposite extreme: its moment of inertia is so large that one quantum of angular momentum costs an imperceptibly small energy. Atomic rotational or orbital steps lie between these extremes, commonly at the scale of electron volts.

Question from the audience. Could an apparently elementary electron still have spatial structure and rotational excitations?[a]

Response. It could only hide them at very high energy. The cost of one unit of angular momentum scales as $1/I$, and $I \sim MR^2$. An exceedingly small radius therefore pushes the first excitation far away on the mass-squared plot. The absence of observed electron excitations means either that the electron is pointlike or that any compositeness scale is much smaller than present experiments can resolve.

[a] Lecture timestamp: 01:42:04.

The opening Regge plot can now be read dynamically. Hadrons reveal accessible rotational and vibrational states because they are extended. The equal steps in their mass-squared spectrum are the signature expected when an oscillator Hamiltonian is identified with M^2. Their approximate linear energy with length is the signature of a flux tube with constant tension. Spectroscopy, scattering duality, world-sheet geometry, and light-cone mechanics are four descriptions of the same organizing idea: an extended relativistic object whose natural internal variable is mass squared.